

Minh-Thanh Nguyen

+353-874-532-647 | <https://scss.tcd.ie/~nguyenm2> | nguyenm2@tcd.ie

EDUCATION

Trinity College Dublin, The University of Dublin <i>Ph.D. in Computer Science</i>	Dublin, Ireland 03.2026 – 02.2030
Hanoi University of Science and Technology (HUST) <i>B.Eng. in Electronics and Telecommunications. GPA: 3.32/4.0 (very good)</i>	Hanoi, Vietnam 09.2021 – 08.2025

PROJECTS

Energy aware resource allocation in Open RAN using multi-agent DRL <i>PhD Student</i>	03.2026 – 02.2030 ICCLab, TCD
- Formulate optimization model for allocating resources in Open RAN with energy considerations. - Build a simulation framework for comparing performance of different algorithms for various Open RAN resource allocation with green energy consumption problems. - Build a multi-agent DRL system which solve the problem of energy aware resource allocation in Open RAN.	
Deploy 5G network slicing testbed <i>Research assistant</i>	01.2025 – 02.2026 ANSALab, HUST
- Deploy and administration a private cloud computing system. - Deploy a 5G networks with basic components with end-to-end slicing capability.	
Efficient resource allocation for 5G network slicing <i>Research assistant</i>	09.2023 – 10.2025 ANSALab, HUST
- Build a simulation framework for comparing performance of different algorithms for various variant of network slice embedding problems. - Formulate optimization model for different variants of network slice embedding problem.	

SKILLS

Programming: Python, C/C++, C#, JavaScripts, Lua, Bash, PowerShell
Technical: OpenStack, Kubernetes, Basic Linux Server Administration
Languages: Vietnamese (native), English (IELTS 7.0)
Writings: Microsoft Word, Microsoft PowerPoint, L^AT_EX

AWARDS

Best Graduation Thesis Award <i>Hanoi University of Science and Technology</i>	Jul. 2025
First Prize at 2025 Student Research Competition <i>Hanoi University of Science and Technology</i>	Jun. 2025
Best Poster Award <i>Asian Internet Engineering Conference 2023, Hanoi</i>	Dec. 2023

PUBLICATIONS

Journals

- [J1] Q.-T. Luu, **M.-T. Nguyen**, T.-A. Do, M. Kieffer, T. H. Nguyen, H. T. Nguyen, and V.-D. Nguyen, “Network slicing with flexible vnf order: A branch-and-bound approach,” *IEEE Transactions on Network and Service Management*, vol. 23, pp. 3805–3818, 2026.
- [J2] Q.-T. Luu, D.-M. Tran, **M.-T. Nguyen**, M. Kieffer, D. T. Hoang, T.-H. Nguyen, H.-T. Nguyen, and V.-D. Nguyen, “Network slice embedding with flexible configurations in 5g networks and beyond,” *IEEE Networking Letters*, pp. 1–1, 2026.

Conferences

- [C1] Q.-T. Luu, **M.-T. Nguyen**, T.-H. Nguyen, M. Kieffer, V.-D. Nguyen, Q.-L. Luu, and T.-T. Nguyen, “Admission Control and Embedding of Network Slices with Flexible VNF Order,” in *2024 20th International Conference on Network and Service Management (CNSM)*, 2024, pp. 1–5.
- [C2] **M.-T. Nguyen**, Q.-T. Luu, T.-H. Nguyen, D.-M. Tran, T.-A. Do, K.-H. Do, and V.-H. Nguyen, “Accelerating Network Slice Embedding with Reinforcement Learning,” in *2024 Tenth International Conference on Communications and Electronics (ICCE)*, 2024, pp. 78–83.

REFERENCES

Asst. Prof. Van-Dinh Nguyen <i>nguyenva@tcd.ie</i>	School of Computer Science and Statistics, Trinity College Dublin O’Reilly Institute, Trinity College Dublin, Dublin 2, Co. Dublin, Ireland
Assoc. Prof. Quang-Trung Luu <i>quangtrung.luu@centralesupelec.fr</i>	CentraleSupélec, Université Paris-Saclay 3 rue Joliot-Curie, 91192 Gif-sur-Yvette, France